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Spatial Biodiversity Modeling Pipeline and Benchmark

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1 Introduction

1.1 Motivation

Biodiversity is defined as "the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems."[1] Biodiversity, in short, is the collection of all living organisms of an ecosystem from plants to animals to fungi. Biodiversity is integral to the functions and stability of ecosystems and plays a pivotal role in maintaining ecosystem functions, recycling nutrients, breaking down pollutants, regulating water systems, carbon storage,stabilizing total biomass within a community, and driving the productivity of an ecosystem (Stork 2009; Cardinale et al. 2012; Gascon et al. 2015). Furthermore biodiversity's significant for human survival is immense, as it provides food, materials, medicine, water security, health, well-being, energy, transportation, climate regulation, and more (Stork 2009; Gascon et al. 2015; Brauman et al. 2020; Kim et al. 2024).

Despite the overwhelming benefits of biodiversity to humanity and ecosystems, biodiversity is declining at an unprecedented rate. Out of all the world's terrestrial wilderness ecosystems approximately 74% of this land area has experienced erosion and approximately 9.6% has been completely lost since the early 1990s (Watson et al. 2016). Some of the most impacted areas of biodiversity loss include marine ecosystems where 66% of the ocean's surface has been significantly impacted by human activities (Díaz et al. 2019). Moreover, it is reasonably likely that most of coral reefs will be lost if global temperatures exceed 2°C (Frieler et al. 2012). Furthermore, human activities threaten the survival of approximately 1 million species, 25% of plants and animals are threatened, and human activities have caused an explosion in the rate of species extinction, which is now estimated to be around 1,000 times the natural background rate (Pimm et al. 2014; Díaz et al. 2019). To make matters worse, conservation efforts are underfunded, with a deficit exceeding half a trillion US dollars annually (Deutz et al. 2020). The acceleration of taxonomic loss and the drastic underfunding of conservation efforts have necessitated a shift towards prioritization of biodiversity conservation and ecological rehabilitation.

Many initiatives around the world are being adopted to combat biodiversity loss and climate change which are intricately linked. These initiatives focus on international cooperation, expanding protected areas, reducing pollution, and incentives for sustainable practices (IPBES 2019; IPCC 2023). Specific policy initiatives such as the 30x30 target-which aims to protect 30% of the world's land area by 2030-represent critical efforts for ecological recovery. If successful, we could see large gains and rebounds in biodiversity. Achieving these goals would have significant impacts including: carbon sequestration, increasing water quality regulation, and expanding capabilities to mitigate nutrient pollution (Zeng, Koh, and Wilcove 2022). For these initiatives to be successful, it is crucial to have accurate and robust models that can predict biodiversity patterns and trends. These models can inform conservation strategies, identify priority areas for protection, and assess the potential impacts of different conservation actions.

At their core, biodiversity models are spatial models that assume biodiversity is influenced by a wide range of environmental variables at a specific spatial location (Ferrier and Guisan 2006). Over the past few decades, a wide range of models have been developed to predict biodiversity patterns and trends. Broadly, biodiversity models can be grouped into three main families: species distribution models (SDMs), which use a bottom-up or "predict first, assemble later" strategy; joint species distribution models (JSDMs), which jointly model species' responses and co-occurrence patterns through an "assemble and predict together" strategy; and community-level models, which use a top-down approach to model biodiversity metrics or community properties directly (Zhang et al. 2019; Pollock et al. 2020). The most common of these models are SDMs, which predict the spatial distribution of individual species based on a wide range of environmental variables (Beery et al. 2021). These individual species predictions can then be aggregated together to predict community level biodiversity metrics (Ferrier and Guisan 2006). Next we have JSDMs, which model multiple species at once using environmental variables and species interactions (Warton et al. 2015). Finally, we have community level models, which model biodiversity as a whole and can predict biodiversity metrics directly without predicting specific species (Pollock et al. 2020).

While each family has their own strengths and weakness and excels in different context it is unlikely one

[1] Convention on Biological Diversity (1993) *Article 2. Use of terms.* Available at: https://www.cbd.int/convention/articles/?a=cbd-02 (Accessed: 8 April 2026).

model will always outperform the others (Ferrier and Guisan 2006; Zhang et al. 2019). But what is clear, is that SDMs are the most researched model family to use for biodiversity prediction. Many studies have focused on improving SDMs, from benchmarks (Valavi et al. 2022) to reviews and evaluations of specific models (Norberg et al. 2019; Waldock et al. 2022), to standardizing best practices for SDMs (Araújo, Anderson, et al. 2019). Similar to SDMs, JSDMs have also been the focus of many studies specifically to combat the neglection of species interactions in SDMs (Wilkinson et al. 2019; Escamilla Molgora et al. 2022). Community level models, on the other hand, are lesser known and underutilized (Nieto-Lugilde et al. 2018). A significant gap remains in our understanding, comparison, and evaluation of SDMs and community-level models, both between these modeling approaches and within the different methods they encompass. This highlights the need for further research and methodological development in community-level biodiversity modeling (Ferrier and Guisan 2006; Nieto-Lugilde et al. 2018; Pollock et al. 2020).

Currently, most community level models are compared within individual studies, across a wide range of datasets, and with varying methodologies, making it difficult to draw general conclusions about their performance and applicability (Cai et al. 2023; Ngor et al. 2023; Baggström et al. 2025). While there have been some reviews of a community level models (Nieto-Lugilde et al. 2018; Mondal and Bhat 2021), there is lack of best practices for this model family which SDMs have (Araújo, Anderson, et al. 2019). This has necessitated the need for a standardized pipeline for community level modeling to allow for more rigorous testing of these models (Pollock et al. 2020). This pipeline would allow for a more systematic evaluation of community level models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a standardized pipeline would allow for a more nuanced evaluation of community-level models and their applicability across different ecological contexts, including comparisons across taxonomic groups, spatial scales, and environmental predictor sets. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

1.2 Research Objectives and Scope

The primary objective of this master's thesis is to develop a standardized pipeline for community level biodiversity modeling and to provide a case study demonstrating the applicability and flexibility of this pipeline. To achieve this objective, first a pipeline is developed that integrates generalized data preprocessing, feature engineering, model training, and evaluation steps for community level modeling. This pipeline is designed to be flexible and adaptable to a wide range of datasets and modeling approaches. Next, a case study is conducted using this pipeline to compare the performance of different community level models on derived biodiversity metrics from the United Kingdom Butterfly Monitoring Scheme (UKBMS) dataset (UK Butterfly Monitoring Scheme 2026) from the Global Information Biodiversity Facility[2]. Models used in this case study will include statistical models, generalized linear models and generalized additive models, and machine learning models, random forests, gradient boosting, and artificial neural networks (Nelder and Wedderburn 1972; T. J. Hastie and Tibshirani 1986; Breiman 2001; Friedman 2001; Lek et al. 1996). These results are then analyzed to provide insights into the strengths and weaknesses of different community level models across different contexts. By applying the pipeline across contexts that vary in taxonomic grouping, spatial scale, environmental predictor sets, and evaluation strategy, this thesis demonstrates its flexibility and general applicability to a range of community-level biodiversity modeling scenarios. This comparative framework also enables a systematic assessment of how different modeling approaches perform under varying analytical conditions, thereby providing insight into their relative strengths, limitations, and appropriate use cases. The key research questions that this master's thesis aims to address are:

1. How do models perform against each other and against an ensemble approach?
2. How do models perform with different strategies of cross validation, specifically random cross validation vs. spatial cross validation?
3. How do models perform in data limited scenarios?

[2]Global Biodiversity Information Facility (GBIF). Available at: https://www.gbif.org/ (Accessed: 26 April 2026).

The main scope of this master's thesis is focused on the development and application of a standardized pipeline for community level biodiversity modeling. While the pipeline is designed to be flexible and adaptable to a wide range of datasets and modeling approaches, the case study conducted in this thesis is limited to the UKBMS dataset and the specific set of modeling approaches described above. The results are therefore intended to illustrate the flexibility and practical utility of the pipeline, while demonstrating how standardized workflows can support more efficient generation of community-level biodiversity insights.

The remainder of this thesis is structured as follows. Section 2 provides background on biodiversity, including how it is defined and measured in Section 2.1, and reviews the main modeling approaches used to predict biodiversity patterns and trends in Section 2.2. Section 3 describes the methodological framework, including the pipeline design in Section 3.1, the case study dataset in Section 3.2, and the models implemented in Section 3.3. Section 4 presents the results of the case study and illustrates the flexibility of the proposed pipeline. Section 5 discusses the implications of these results for biodiversity modeling and conservation, while Section 5.2 addresses relevant ethical considerations. Section 6 outlines potential future directions and applications for research in this area. Finally, Section 7 concludes the thesis by summarising the key findings and their implications for biodiversity modeling and conservation.

2 Background

This section covers biodiversity and how it is measured in Section 2.1 and reviews the model techniques used to predict biodiversity patterns and trends in Section 2.2. This background section provides the necessary context for understanding the methods and results presented in this thesis. It will also highlight the differences between the modeling approaches as well as the different assumptions and limitations of these models.

2.1 Biodiversity and how to measure it

Biodiversity is a complex and multifaceted concept that encompasses the variety of life on Earth, including the diversity of species, genetic diversity within species, and the diversity of ecosystems (Stork 2009).

2.2 Spatial Biodiversity Modeling

Species Distribution Models (SDMs) are a class of models that predict the occurrence (presence or absence) or abundance of individual species at unique geographic locations. SDMs utilize a wide range of variables such as temperature, precipitation, land cover, and human presence to make predictions on the distribution of individual species by establishing relationships between these variables and species occurrences (Beery et al. 2021). While the first SDMs were developed in the early 20th century with distributions of the Chestnut-backed Chickadee (Grinnell 1904), the application of these models has been extended beyond just generating range maps (Beery et al. 2021). They now encompass the pursuit of insights into the causal drivers of species distributions, land use management planning, conservation planning (Elith and Leathwick 2009), revealing biological patterns, and even helping with the discovery of sites for the reintroduction of threatened species (Sinclair, White, and Newell 2010). The versatility of these models has led to their widespread use in ecology and conservation biology especially in the past few decades.

The utility of SDMs was first expanded by the development of the Generalized Linear Model (GLM) and Generalized Additive Model (GAM) frameworks in the late 1970s and 1980s. Before these developments SDMs were primarily ecological models based on conceptual or expert knowledge of species distributions. One of the largest challenges of statistical SDMs models was the assumption of Gaussian error distributions for linear models. This assumption is often violated in ecological data. For example, count data follow a Poisson distribution (Guisan, Edwards, and T. Hastie 2002). The main breakthrough was first with GLMs. GLMs are able to model non-Gaussian-distributed data; specifically, any distribution in the exponential family can be modeled with a GLM (Nelder and Wedderburn 1972). GAMs then further extended the capabilities of GLMs by allowing for nonlinear relationships between the predictors and the response variable (T. J. Hastie and Tibshirani 1986). However, even with the advance of GLMs and GAMs to model non-Gaussian data and nonlinear relationships, SDMs still would struggle when there would be a dependency structure in the data and violated independent and identically distribution (i.i.d.) assumptions d(Guisan, Edwards, and T. Hastie 2002; Beery et al. 2021). To address these issues and the need to model more complex relationships machine learning models including random forests, boosted regression trees, gradient boosting, neural networks, and ensemble methods have been increasingly used (Norberg et al. 2019; Beery et al. 2021; Valavi et al. 2022; Waldock et al. 2022; Özkan Tümer et al. 2026). These methods can capture complex nonlinear relationships and interactions between predictors and the response variable, and they can handle large datasets with many predictor variables.

Concurrently, large advancements in the organization and distribution of biodiversity data through organizations such as the Global Biodiversity Information Facility (GBIF)[3] and national databases has reduced barriers to data access (Özkan Tümer et al. 2026). In addition to open biodiversity datasets, the quality of biodiversity datasets has continued to improve the quality of species data (Araújo, Anderson, et al. 2019). Moreover, advancements in geographical information systems (GIS) have allowed for the integration of spatial data into SDMs. This spatial data can be of climatic, environmental, topographic,

[3] Global Biodiversity Information Facility (GBIF). Available at: https://www.gbif.org/ (Accessed: 19 April 2026).

or anthropogenic nature across larger areas and finer temporal scales (Elith and Leathwick 2009; Waltari et al. 2014; Özkan Tümer et al. 2026). These advancements alongside with the development of new modeling techniques have further driven the advancement of SDMs with dozens of models being developed across hundreds of papers in the past few decades (Valavi et al. 2022; Özkan Tümer et al. 2026).

SDMs have been the dominant modeling family for species prediction for several decades and have been widely used in ecology and conservation biology and for good reason. They excel at predicting the distribution of individual species and contribute to our understanding of species niches (Sinclair, White, and Newell 2010). However, they are not without their shortcomings. Firstly, and the most obvious when it comes to measuring biodiversity is that SDMs are designed to predict the distribution of individual species (Ferrier and Guisan 2006), and most SDMs do not capture biotic interactions between species, which can be a critical driver of species distributions and community composition (Waldock et al. 2022). If we want to derive biodiversity metrics from SDMs, we would have to fit a separate SDM for each species and then aggregate the results together in a "predict first, assemble later" or bottom-up approach. This can be in itself can be very time-consuming not only to building the models themselves but also to gather all the species data and environmental data across spatial and temporal scales (Ferrier and Guisan 2006). Compounding this issue is the fact that most species data is still presence only data which would make biodiversity metric calculations aside from species richness difficult (Valavi et al. 2022).

Furthermore, all SDMs are not created equally in varying environmental contexts. In some contexts such as measuring abundance, a random forest may be the best model (Waldock et al. 2022), while in a presence only context an ensemble based model may be the best (Valavi et al. 2022). Other studies have found large variations in model performance and one model may not rule them all and that the best model can vary across species and contexts (Norberg et al. 2019). This is further exacerbated by the fact that hyperparameter tuning or parameterization and feature selection models can also have a large effect on the model performance (Araújo and Guisan 2006; Beery et al. 2021). So, if we wanted to use SDMs to predict biodiversity metrics, not only would we have to fit a separate SDM for each species but it is likely we would have to fit multiple models, and even multiple versions of the same model, for each species to find the best model for each species. SDMs also struggle to predict rare or infrequent species which can account for a large portion of species richness metrics (Ferrier and Guisan 2006; Norberg et al. 2019). Many studies also note how transferability or extrapolation to unsampled sites is a common challenge for SDMs and is in part due to autocorrelation of environmental and spatial variables (Norberg et al. 2019; A. Lee-Yaw et al. 2022). Finally, SDMs tend to be biased at the extremes of abundance distributions, frequently overestimating low-abundance sites while underestimating sites with high abundance (Waldock et al. 2022).

We can see that SDMs have advanced from ecological models based on expert knowledge (Grinnell 1904), to parametric statistical models (Guisan, Edwards, and T. Hastie 2002), to machine learning models (Özkan Tümer et al. 2026). The advancement of SDMs has concurrently been driven by advancements in quality of species data as a response variable and (Araújo, Anderson, et al. 2019), environmental data as predictor variables (Beery et al. 2021). SDMs can do well when it comes to prediction occurrences of individual species (A. Lee-Yaw et al. 2022); however, the shortcomings of SDMs as a model family when it comes to modeling biodiversity metrics are significant. While aggregating multiple SDMs can be used to predict biodiversity metrics, other methods such as joint SDMs (JSMDs) have shown to be more effective at predicting biodiversity metrics (Norberg et al. 2019).

3 Methods/setup

3.1 Pipeline for Community Level Modeling

Rename section to something with spatial biodiversity pipeline Was built on Jakob's pipeline and cite.

3.2 Data

3.3 Models

Go more into the details of the statistical methods?? Focus more on top down methods and the ML methods

4 Results/ Case Study

5 Discussion

5.1 Findings

5.2 Ethical Discussion

6 Future Work/ Applications

Future Work includes a full comprehensive benchmark of community level models across a wide range of datasets and models. The integration with GBIF should should allow for easy facilitation of this. This would allow for a more systematic evaluation of community level models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a comprehensive benchmark would allow for a more nuanced evaluation of community-level models and their applicability across different ecological contexts, including comparisons across taxonomic groups, spatial scales, and environmental predictor sets. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

A comprehensive benchmark or review of both community level models and SDMs and JSDMs.

Integration of better output comparison and visualization tools into the pipeline. This would allow for easier interpretation of model results and better communication of findings to stakeholders and policymakers.

More input types

7 Conclusion

A Appendix A: Github Repository

Includes the instructions there

B Appendix C: Additional Results

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